Manual

Ranges of Coverage and Material Availability

MPL-RMV 8.2

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Contents

1 Ranges of Coverage and Material Availability

Summary

Purpose

The function package "ranges of coverage and material availability" complements the function packages "material and production logistics" as well as "tracking & tracing" by functions to represent the material flow and monitor ranges of coverage in production.

Machines/workplaces defined in the system are connected with material buffers defined in the system and grouped to production levels.

These production levels are used to define delivery relationships and, as a result, to represent the material flow in production as basis for estimations of the range of coverage.

Implementation Considerations

The function package "ranges of coverage and material availability" is used if you

- would like to monitor the material flow and range of coverage of material;
- would like to make statements on the current or expected material availability, in particular, for WIP material.

Integration

The function package "ranges of coverage and material availability" uses the machines/workplaces defined in the system as well as the material buffers created in the system.

The orders/operations existing in the system are referred for considerations relating to the range of coverage.

Features

- Configuration of production levels
    - Configuration of production levels to which the machines and material buffers are assigned from which the machines of the corresponding level may withdraw material.
- Configuration of supply relationships
    - Representation of supply relationships between the production levels. In this context, one production level may have several subsequent levels or preceding levels.
- Checking of material availability
    - Checking of material availability for an existing situation (e.g. production level)
- Estimation of range of coverage

- Estimation of the range of coverage for materials transferred between two production levels

2 Range of Coverage Analysis and Material Availability

Usage

For the purpose of analyzing material coverage ranges and/or material availability, so-called production levels are defined. In this process, various machines as well as material buffers from where the machines of a specific production level may retrieve material are assigned to a specific production level.

In addition, the supply relationship between production levels is modeled. A production level may have several subsequent and/or preceding levels. In this supply relationship, a minimum coverage range is defined.

Prerequisites for Configuration

- Definition of production levels
- Assignment of machines/workplaces and consistently their material buffer for production levels.
- Assignment of additional material buffers for production levels.
- Definition of supply relationships between production levels; in this regard, a preceding level is considered as supplying, a subsequent level is considered as consuming.
- A preceding level may have several subsequent levels.
- A subsequent level may have several preceding levels.

Configuration of Production Levels

Define various production levels and create these production levels in the system.

Assign the related machines to the created production levels (production level assignment). By assigning a machine, its allocated material buffers will automatically be assigned, too. It is not possible to delete these material buffers explicitly, but they will be deleted automatically upon deletion of the assignment of the related machine.

Assign the related material buffers to the created production levels (production level assignment).

Configuration of Supply Relationships

Define various supply relationships and create these supply relationships in the system by assigning one preceding level and one subsequent level from the defined production levels in the system to each supply relationship.

For each preceding level (predecessor production level), several entries can be created for several subsequent levels (consumer/successor).

For each subsequent level (consumer/successor production level), several entries can be created for several preceding levels (predecessor).

3 MBL Range of Coverage Analysis

Overview

The range of coverage analysis allows the user to recognize how much time is left until a specific material for a production level is consumed and/or for how long there will be enough material available for production.

This means that this function analyzes the range of coverage of a material transferred between two consecutive production levels (preceding level and subsequent level).

The range of coverage computation is requested for a *production level* considered as consuming in this context. With regard to the analysis, all supply levels/production levels assigned to this level as preceding levels are then considered.

If these, in turn, are supply levels to other levels, the latter are also included for the purpose of a holistic analysis.

In this process, the following materials from the batch stock are included:

- Material of the selected production level
  - Material in the configured input buffer of the production level
  - Material currently being used as component in the running operation (to be found through BOM)
- Material from the preceding levels to the selected production level
  - Material currently located in configured output buffers of preceding levels
  - Material currently produced in the preceding levels of the running operation

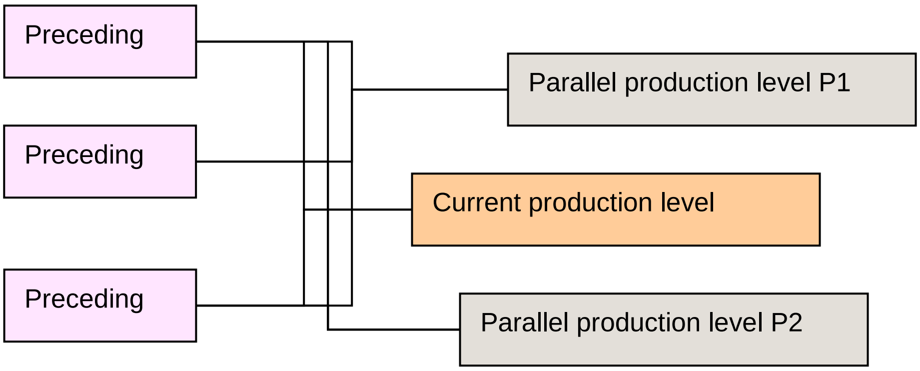
For each material, the number of supplying as well as consuming machines is indicated.

The current stock of material buffers to be analyzed is considered as initial stock. In this regard, batches currently logged on to consuming machines are also considered.

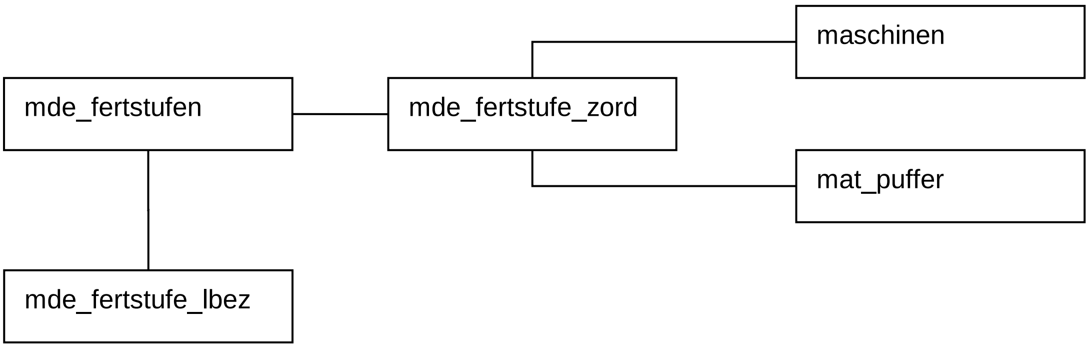
For calculating the range of coverage, the criteria listed below are therefore relevant:

- The consuming OPs and their target cycle result in a demand per unit of time (e.g. component A = 100 pcs/h are consumed).
- The supplying OPs and their target cycle result in a stock per unit of time (e.g. component A = 100 pcs/h are produced).

Connections



Configuration Data Model



Selection

- Production level
  The range of coverage computation is requested for a production level considered as consuming in this context. With regard to the analysis, all production levels assigned to this level as preceding levels are then considered. If these, in turn, are supply levels to other levels, these other, retrieving production levels are also included for the purpose of a holistic analysis.

- Incl. materials which are (currently) only supplied/produced
  This option allows for extending the analysis to also include materials which are currently only supplied and/or only produced.

Display

The range of coverage is displayed in hours as well as a trend (arrows/signs/symbols). The display is updated automatically and/or cyclically every 180 seconds.

The range of coverage analysis grid shows the data listed below:

Designation	Description, source
Material	
Material	• In producing operations, the material corresponds to the materials in the component list • In supplying operations, the material corresponds to the article ID
Designation	Material designation from batch stock for selected material numbers
Stock	Quantity: • Batches to be selected from batch stock with relevant material number • Selected batches are always in the free/running status (in buffer/posted on operation). Processed or locked batches are not used. • If the remaining quantity is < 0, these batches are not considered.
Unit	Unit from batch stock for selected quantity
Range of coverage	
Range of coverage	Range of coverage in hours (decimal); default: not visible
Previous range of coverage	Range of coverage in hours (decimal); default: not visible
Trend	In addition, the trend of the range of coverage (change in comparison to previous state) is indicated in front of each bar by an appropriate symbol. - increasing ➚ - decreasing ➘ - unchanged ➔
1	Hour; bar chart
:	Hour; bar chart
16	Hour; bar chart
17	Hour, bar chart; default: not visible
:	Hour, bar chart; default: not visible
24	Hour, bar chart; default: not visible
Retrieving machines	
Total	Machines within the supply relationships of the current production level and with currently running operations using this material.
In production	Machines within the supply relationships of the current production level and with currently running operations using this material and operating in the *Production* status.
Supplying machines	
Total	Machines of other supply relationships with currently running operations using this material.
In production	Machines within the supply relationships of the current production level and with currently running operations producing this material.

Data Acquisition

Data are acquired for a production level considered as consuming in this context. With regard to the analysis, all production levels assigned to this level as preceding levels are then considered. If these, in turn, are supply levels to other levels, these (other levels) are also included for the purpose of a holistic analysis.

Parameter Entry

FSTUFE : selected production level

PROD=J : incl. materials (currently) only supplied/produced
→ no range of coverage is calculated for these (range of coverage = 0)

Value Return

Column	Comment/Layout
ARTIKEL	Material
BEZ	Material designation
BESTAND	Calculated material stock
EINHEIT	Quantity unit of material
REICHW	Range of coverage: calculated value
ANZ_ENT_MNR	Number of retrieving machines
ANZ_ENT_MNR_PROD	Number of retrieving machines in production
ANT_ENT_MNR_ANDERE	Number of retrieving machines from other production levels
ANZ_PROD_MNR	Number of producing machines
ANZ_PROD_MNR_PROD	Number of producing machines in production
MIN_REICHW	Minimum range of coverage of material; always 0!

The following criteria are relevant for the range of coverage:

- The consuming OPs (current production level and production levels running in parallel, if any) and their cycle result in a demand per unit of time (e.g. component A = 100 pcs/h are consumed).

- The supplying OPs (preceding production levels) and their cycle result in a stock per unit of time (e.g. component A = 100 pcs/h are produced).

The initial stock is the current material stock from assigned material buffers of preceding levels. In this regard, batches currently logged on to consuming machines must also be considered.

Procedure

Identification of all machines to be considered

Recommendation: Enter the machines in a temporary table.

- Machines of the current production level

- Machines of production levels preceding the current production level (preceding levels)

- Machines of other production levels following the preceding levels (quasi existing in parallel 'P' to the current production level)

Identification of materials used on machines of the current level

(For these, the range of coverage has to be determined, i.e. how long will the current stock last.)

For each currently running operation (AUFTRAG_STATUS.PROD_KENN = 'L') and its consuming material components (MLST_HY.KENNZ 'M', T') in production on machines of the current production level, the required quantity per hour is calculated.

The basis for this is the target cycle entered for the operation (= speed at which the article is produced; AUFTRAGS_BESTAND.SOLL_DAUER/SOLL_TEIL) as well as the input quantity of the supplied material components (MLST_HY.SOLL_MENGE).

Time required for producing 1 piece = $\dfrac{ab.soll\_dauer \,/\, ab.soll\_teil}{1000}$ [s]

Time required for a material to be used = Timeperpcs = $\dfrac{\left(\dfrac{ab.soll\_dauer \,/\, ab.soll\_teil}{1000}\right)}{mlst\_hy.soll\_menge} \left[\dfrac{s}{UOM}\right]$

Required quantity per hour for a material to be used = $\dfrac{3600\ [s]}{TimePerItem \left[\dfrac{s}{UOM}\right]}$

Since an article may be processed at different target cycle times in different orders/operations, the individual values are determined as an added up total.

When calculating the range of coverage, the OPs logged on to machines of levels running in parallel (P1, P2) must also be considered. For this reason, the "Required quantity per hour for a material to be used" must be identified for each of them, too.

Identification of initial stocks

The initial stock is the current material stock (LOS_BESTAND.RESTMENGE) from assigned material buffers of preceding levels. The batches with "L" and "F" status (LOS_BESTAND.STATUS) whose residual quantity is > 0 are considered.

Identification of materials produced on machines of preceding levels

For materials produced in the preceding level, the speed at which the current operations (AUFTRAG_STATUS.PROD_KENN = 'L') produce the materials on machines of preceding levels (AUFTRAGS_BESTAND.ARTIKEL) is calculated.

Time required for producing 1 piece = Timeperpcs = $\frac{ab.soll\_dauer / ab.soll\_teil}{1000}$ [s]

Quantity produced per hour = $\frac{3600\ [s]}{TimePerItem\left[\frac{s}{UOM}\right]}$

Calculation of range of coverage

The range of coverage of each material [in hours] is determined from the ratio of the values described above:

Range of coverage [h] =

$$\frac{LOS\_BESTAND.RESTMENGE}{\frac{3600.0}{TimePerItem(ANR1)} + \frac{3600.0}{TimPerItem(ANR2)} - \frac{3600.0}{TimePerItem(ANRn)} + \ ....}$$

\+ producing OPs (range of coverage increases)

\- consuming OPs of current level (range of coverage decreases)

\- consuming OPs of level(s) running in parallel (range of coverage decreases even more)

If the option "Incl. materials which are (currently) only supplied/produced" is set, not only the machines used in the current production level but also the materials only produced in the preceding level, are identified. For these, however, only the stock, but not the range of coverage will be calculated.

Number of retrieving machines

Number of retrieving machines (i.e. of the production level to be considered) currently using this article (based on component list of active order).

Number of retrieving machines (i.e. of the production level to be considered) currently in PRODUCTION (STOER_TABELLE.PROD_KENN = 'P'') and using this article (based on component list of active order).

Number of retrieving machines from other production levels

Number of retrieving machines in production levels which, in relation to preceding levels, are downstream and which currently use the articles (based on component list of active order) also used on machines of the current production level.

Number of supplying machines

Number of supplying machines (i.e. of the preceding levels) currently producing these articles.

Number of supplying machines (i.e. of the preceding levels) currently producing these articles and in PRODUCTION.

4 Production Levels

1.1 Summary

Menu	Master data → Material → Production Levels
Transaction code	plev
Function authorization	plev

Utilization

The function is used to create or modify production levels within the system and to assign machines or material buffers for each production level.

Integration

The creation of production levels allows for machines and material buffers, which may be used in different evaluations/reports e.g. in the estimation of the range of coverage, to be summarized individually.

The application "production levels" is divided into two areas:

- Production levels and their editing functions
  A production level may be selected, added or edited in this dialog.
- Assignment functions and their relevant editing functions
  This screen shows the machines and material buffers assigned to the selected production level. Further machines and material buffers are also added or edited.

Selection criteria

The application provides the following selection criteria:

Production level

The production level may be selected or entered.

Material buffer

The material buffer can be selected or entered.

Machine

The machine may be selected or entered.

Field descriptions "production levels"

Production level

The production level may be selected or entered.

Description

Designation of the production level

Editing functions

The following dialog opens to edit a data record:

**Add**

Inserts a new production level

**Edit**

Edits an existing production level

**Delete**

Deletes the selected or several selected production levels

"Production levels assignment" detail application

The "assignment" detail application shows the below-mentioned data for each production level:

Category, type

These assignments are possible:

- Machine
- Material buffer

Name

Subject to the assignment, this field shows the machine number or the material buffer.

Description

Comment field / text field

Company

Abbreviated company name

Cost center

Affected cost center

Site / company

Affected site

Editor

Last editor of the data record

Last modification

Last modification to the data record.

The documentation dealing with the assignment of production levels describes the editing details for the selected production levels and, as a result, the detail application "assignment".

5 Assignment of Production Levels

Overview

Menu	Master data → Material → Production levels → Assignment
Transaction code	asplev
Function authorization	asplev

Usage

This function is used to create or modify the assignment of production levels in the system.

Integration

Forming production levels enables an individual summary of machines and material buffers that can be used in various evaluations, e.g. in the analysis of the range of coverage.

Requirement

The production levels must be defined in the system.

Selection criteria

The following selection criteria are available in the application:

Production level

Selection or input of the production level

Material buffer

Selection or input of the material buffer

Machine

Selection or input of the machine

Editing functions

The following dialog opens to edit a data record:

Add

Adds a new assignment.

Delete

Deletes the selected or several selected assignments.

Field descriptions "production levels - assignment"

Once the editing dialog is opened, the below-mentioned fields are available:

Production level

If a production level is selected, only this one will be displayed. Wildcard characters may be used in this selection option.

Type

Machine: A machine is assigned to the production level.

Material buffer: A material buffer is assigned to the production level.

Machine

Machine which is to be assigned to the production level.

If one or several machines with similar name (input of wildcard characters) is/are selected, the display of production levels that might be selected is restricted to the production levels that this machine is assigned to.

Material buffer

Material buffer that is to be assigned to the production level.

If one or several material buffers with similar name (input of wildcard characters) is/are selected, the display of production levels that might be selected is restricted to the production levels that this material buffer is assigned to.

6 Closed Loops / Supply Relationships

Summary

Menu	Master data → Material → Supply relationships
Transaction code	intsc
Function authorization	intsc

Usage

By forming closed loops/supply relationships, the successor and predecessor levels of defined production levels can be defined. One production level can have several successor and predecessor levels.

Integration

The closed loops/supply relationships are used in the evaluation of ranges as well as in the e-kanban process as base data.

Requirements

The production levels must already be created in the system.

Selection criteria

The following selection criteria are available in the application:

Preceding level/supply - production level

Displays all supply relationships with the selected production level

Preceding level/supply - description

Displays all supply relationships with the selected description

Subsequent level/consumption - production level

Displays all supply relationships with the selected production level

Subsequent level/consumption - description

Displays all supply relationships with the selected description

The user authorization "kov" is required for displaying the below-mentioned fields.

Material number

Shows all closed loops/supply relationships matching the entered material number.

Closed loop/supply relationship ID

Shows the closed loop/supply relationship matching the entered ID that is specific within the system.

Field descriptions

"General" index tab:

Preceding level/supply

Selected preceding level/supply of a closed loop

If e-kanban is in use the preceding level/supply (place of production) represents the starting point of the closed loop for a kanban object. The supply is chosen from the production levels created in the system.

In case the supply is a supermarket, warehouse or intermediate production buffer, it has to be created as material buffer within MES and is chosen from the production levels (the material buffers are assigned to).

If e-kanban is in use, it is not possible to directly assign a machine group as the supply or a relevant material buffer for an entire machine group. But in fact the same material buffer is entered as common material buffer of the entire machine group for all affected machines pertaining to a machine group. This common material buffer is then selected as the supply.

Subsequent level/consumption of the closed loop

Selected subsequent level/consumption of a closed loop.

If e-kanban is in use the consumption represents the end point (place of consumption) of the closed loop for a kanban object. The consumption is chosen from a production level created in the system.

In case the consumption is a supermarket or intermediate production buffer, it has to be created as material buffer within MES and is chosen from the production levels (the material buffers are assigned to).

If e-kanban is in use, it is not possible to directly assign a machine group as the consumption or a relevant material buffer for an entire machine group. But in fact a common material buffer (supplying the whole machine group) is entered or selected as preceding machine buffer for all affected machines pertaining to a machine group.

The user authorization "kov" is required for displaying the below-mentioned fields.

Material number

Material number of the kanban article for which the closed loop applies. Several closed loops (n entries) can be defined for a material.

Closed loop/supply relationship ID

Specifically identifies the closed loop/supply relationship. A supply relationship can only be created once within the system.

"Kanban" index tab:

Number of KBN in circulation

Total number of all kanban objects circulating within this closed loop.

Minimum stock level of empty kanbans (start "yellow" zone)

The minimum stock level of empty kanbans represents the upper stock limit of empty kanban containers that may reached before, for example, production (replenishment) is triggered automatically and a changeable, planned kanban order is generated initially. Hence, the minimum stock level of empty kanbans represents the limit between the "green" and "yellow" zone (status lights at the terminal). The minimum stock level needs to be configured for the display in:

- Electronic Kanban Board (AIP)

Maximum stock level of empty kanbans (start "red“ zone)

The maximum stock level of empty kanbans represents the lower stock limit of empty kanban containers that may be reached. If this limit is exceeded, the planned kanban order, for example, automatically becomes an unchangeable, fixed kanban order. Hence, the maximum stock level of empty kanbans represents the limit between the "yellow" and "red" zone (status lights at the terminal). The maximum stock level needs to be configured for the display in:

- Electronic Kanban Board (AIP)

Planned KBN quantity

Planned, defined quantity of kanban objects (contents).

7 Range of Coverage Analysis and Material Availability

Overview

Menu	Material management → Inventory management → Estimation of range of coverage
Transaction code	roc
Function authorization	roc

Usage

This function analyzes the range of coverage of the material transferred between two consecutive production levels (preceding level and subsequent level).

For generating the list, materials are considered which are

- currently produced in the preceding level
- used as components in the subsequent level

For each material, the number of supplying as well as retrieving machines is now indicated. By means of option buttons, only the supplying or only the consuming machines can be considered.

The following criteria are relevant for the range of coverage:

- The consuming OPs and their target cycle result in a demand per unit of time.
- The supplying OPs and their target cycle result in a demand increase per unit of time.

The current stock of material buffers to be analyzed is considered as initial stock. In this regard, batches currently logged on to consuming machines are also considered.

Integration

For the purpose of analyzing material coverage ranges and/or material availability, so-called production levels are defined. In this process, various machines as well as material buffers from where the machines of a specific production level may retrieve material are allocated to a specific production level.

In addition, the supply relationship between production levels is shaped. A production level may have several subsequent and/or preceding levels. In this supply relationship, a minimum coverage range is defined.

The range of coverage computation is requested for a production level considered as consuming in this context. With regard to the analysis, all supply levels/production levels assigned to this level as preceding levels are then considered. If these, in turn, are supply levels to other levels, the latter are also included for the purpose of a holistic analysis.

Prerequisite

The production levels and supply relationships must exist in the system.

Selection criteria

Production levels

The range of coverage computation is requested for a production level considered as retrieving in this context. With regard to the analysis, all supply levels/production levels assigned to this level as preceding levels are then considered. If these, in turn, are supply levels to other levels, these other, retrieving production levels are also included for the purpose of a holistic analysis.

Incl. materials (currently) only supplied/produced

This option allows for extending the analysis to also include materials which are currently only supplied and/or only produced.

Display

The range of coverage is shown in hours in the screen. For a cyclical display, the data are updated every 180 seconds. The following data are displayed:

Material

The material corresponds to the materials in the component list in producing operations and to the article ID in supplying operations, respectively.

Material designation

Material designation

Number of retrieving machines (total)

Machines within the supply relationships of the current production level and with currently running operations using this material.

In production

Machines within the supply relationships of the current production level and with currently running operations using this material and operating in the *Production* status.

Number of supplying machines (total)

Machines within the supply relationships of the current production level and with currently running operations producing this material.

In production

Machines within the supply relationships of the current production level and with currently running operations producing this material and operating in the *Production* status.

Range of coverage

The range of coverage in [HH:MM:SS] is computed as follows:

In the first step for each operation and material component, the quantity per hour is computed. The basis for this is the target cycle entered for the operation (= speed at which the article is produced), as well as the input quantity of the included material components.

Since an article may be processed at different target cycle times in different orders/operations, the individual values are added up.

The basis for computing the range of coverage of a material is the remaining quantity entered in the batch stock (with material number). The batches considered are those in the material buffers of the immediately preceding level (acc. to configuration). Only batches in the F (free) and L (running) status are considered.

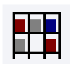
The range of coverage [in hours] is determined from the ratio of the values described above:
- Remaining quantity acc. to batch stock
- Required quantity per hour.

For materials which are indicated in the list (because they are currently produced on a supplying machine), but which are not retrieved (because no retrieving order/operation is active), the range of coverage is not computed.
For materials which are more rapidly produced than consumed, the range of coverage is also infinite.
In these cases, the "infinite" symbol is displayed in the "Trend" column.

Trend

In addition, the trend of the range of coverage (change in comparison to previous state) is indicated in front of each bar by an appropriate symbol.
- increasing ↗
- decreasing ↘
- unchanged →
- infinite ∞

On the first data request, the "unchanged" symbol is displayed.

In the range of coverage analysis, the  function can be used to switch to the "Stock overview" in order to display the stock overview of the previously selected material there.

Each of the materials (batches) and material components must be defined in the same quantity unit; there is no internal quantity conversion.